

THE MYSTERY OF FIRE



What does a flame need to keep burning?



AIR IS EVERYWHERE!

You can't see it — but it's there.

WHOOSH →

WHAT IS BURNING?

HEAT

FUEL

OXYGEN



COMBUSTION!



HEAT + LIGHT



LET'S TEST IT!



SCIENCE LAB TIME!

Candle · Jar · Lid · Clay · Board · Lighter (teacher only)

EXPERIMENT SETUP



1. Candle ready

2. Jar moves over

3. Air around flame

Watch how the jar traps the air around the flame!

WHAT DO YOU THINK WILL HAPPEN?



A STRONG FLAME

B SMALLER FLAME

C FLAME OUT

THE CLOSED JAR



THE AIR
IS TRAPPED
INSIDE

WATCH CAREFULLY!



BIG FLAME → smaller → tiny → OUT

Oxygen particles decrease at the same time!

WHY DID THE FLAME GO OUT?

Oxygen is used up as the candle burns.

In a sealed jar, there is only a limited amount of oxygen.

AT THE START

Plenty of oxygen particles.
The candle burns bright and tall.



LOTS OF OXYGEN

LATER

Oxygen particles are decreasing.
The flame gets smaller.



LESS OXYGEN



When the candle burns, it uses up oxygen.

In a sealed jar, the oxygen runs low, the flame shrinks, and eventually the candle goes out.

1. Oxygen near the flame

2. Oxygen gets used up

3. Flame gets weaker

4. Flame goes out

WHAT IF FRESH AIR CAN ENTER?



Gap at the bottom

Fresh air can flow in

WHOOSH!

An opening lets new air reach the flame.

FRESH AIR ENTERS!



FRESH AIR ↓

OXYGEN →

WARM AIR ↑

IT KEEPS BURNING!



Fresh air enters

Oxygen reaches flame

Flame grows strong!

AHA! Fresh air keeps supplying oxygen.

CLOSED vs OPEN



Air runs out → OUT

Air renewed → BURNS

WHY DOES THIS HAPPEN?

HEAT

+

FUEL

+

OXYGEN

↓ COMBUSTION ↓

HEAT + LIGHT



AIR HELPS FIRE KEEP BURNING



**FRESH AIR
KEEPS THE
FLAME
ALIVE**

Fresh air surrounds the flame — and keeps it burning.

REAL LIFE!



Candle • Fireplace • Stove • Engine — each needs air!

SCIENCE CHALLENGE!

WHICH ONE WILL KEEP BURNING?



A Closed jar

B Open air

Answer: B — fresh air keeps coming in!

REMEMBER!



1. BURNING NEEDS OXYGEN

2. FRESH AIR MUST BE SUPPLIED

3. WITHOUT FRESH AIR, THE FLAME GOES OUT